

RGPV Machine Learning – Detailed Exam Answers

Format: Definition (2–5 lines), Diagram, Explanation, Example, Advantages, Applications and Conclusion.

Linear Regression

Definition: Linear Regression is a supervised learning algorithm used to predict continuous values by establishing a linear relationship between input variables and output variables.

Diagram:

Input Features → Best Fit Line ($Y = mX + c$) → Predicted Value

Explanation:

- Finds the relationship between dependent and independent variables.
- Uses a cost function to measure prediction error.
- Parameters are optimized using gradient descent.
- Performance metrics include MAE, MSE and R^2 score.

Example: Predicting house prices from area and location.

Advantages: Simple, interpretable and computationally efficient.

Applications: Sales forecasting, demand prediction and risk analysis.

Conclusion: Linear Regression is the foundation of predictive analytics for continuous data.

Perceptron Model

Definition: A Perceptron is a single-layer neural network used for binary classification problems.

Diagram:

$x_1, x_2, \dots, x_n \rightarrow \Sigma(wx+b) \rightarrow \text{Activation Function} \rightarrow \text{Output}$

Explanation:

- Inputs are multiplied by weights.
- Bias shifts the decision boundary.
- Activation function generates final output.
- Weights are updated using learning rules.

Example: Classifying emails as spam or not spam.

Advantages: Easy to implement and fast to train.

Applications: Pattern recognition and simple classification tasks.

Conclusion: Perceptron is the basic building block of deep neural networks.

Gradient Descent

Definition: Gradient Descent is an optimization algorithm that minimizes the loss function by iteratively updating model parameters.

Diagram:

Initialize Parameters → Compute Gradient → Update Weights → Repeat

Explanation:

- Moves in the negative gradient direction.
- Learning rate controls update size.
- Types: Batch, Stochastic and Mini-batch.
- Stops when minimum error is achieved.

Example: Optimizing weights in linear regression.

Advantages: Efficient and widely applicable.

Applications: Regression, neural networks and logistic regression.

Conclusion: Gradient Descent enables models to learn optimal parameters.

Backpropagation

Definition: Backpropagation trains neural networks by propagating prediction errors backward and updating weights.

Diagram:

Input → Hidden Layer → Output → Loss ← Backward Error Flow

Explanation:

- Perform forward propagation.
- Calculate loss.
- Use chain rule to compute gradients.
- Update weights using gradient descent.

Example: Handwritten digit recognition.

Advantages: Efficient training of deep networks.

Applications: Image recognition, NLP and speech processing.

Conclusion: Backpropagation is the core learning mechanism of deep learning.

CNN Architecture

Definition: A Convolutional Neural Network is a deep learning model designed for image and visual data processing.

Diagram:

Image → Convolution → ReLU → Pooling → Fully Connected → Output

Explanation:

- Convolution extracts features.

- Pooling reduces dimensions.
- ReLU introduces non-linearity.
- Fully connected layer performs classification.

Example: Face recognition systems.

Advantages: Automatic feature extraction with high accuracy.

Applications: Medical imaging, self-driving cars and object detection.

Conclusion: CNNs revolutionized computer vision applications.

Support Vector Machine

Definition: SVM is a supervised learning algorithm that separates classes using an optimal hyperplane with maximum margin.

Diagram:

Class A ● ● | Margin | Hyperplane | Margin | ■ ■ Class B

Explanation:

- Works for linear and non-linear data.
- Uses kernel functions.
- Maximizes classification margin.
- Effective in high-dimensional spaces.

Example: Cancer diagnosis classification.

Advantages: High accuracy and effective with small datasets.

Applications: Text classification and image recognition.

Conclusion: SVM provides robust classification performance.

LSTM vs GRU

Definition: LSTM and GRU are recurrent neural network variants designed to solve the vanishing gradient problem.

Diagram:

LSTM: Input Gate + Forget Gate + Output Gate | GRU: Update Gate + Reset Gate

Explanation:

- LSTM has more parameters.
- GRU is computationally faster.
- Both handle long-term dependencies.
- Used for sequence modeling.

Example: Language translation and sentiment analysis.

Advantages: Captures sequential information effectively.

Applications: Chatbots, speech recognition and forecasting.

Conclusion: Choose LSTM for complex tasks and GRU for faster training.

Q-Learning vs SARSA

Definition: Q-Learning and SARSA are model-free reinforcement learning algorithms that learn action values.

Diagram:

State → Action → Reward → Update Q-value

Explanation:

- Q-Learning is off-policy.
- SARSA is on-policy.
- Both use Bellman updates.
- Q-Learning is more exploratory.

Example: Game playing agents.

Advantages: Learns optimal actions without environment models.

Applications: Robotics and autonomous systems.

Conclusion: Both algorithms help agents learn through interaction.